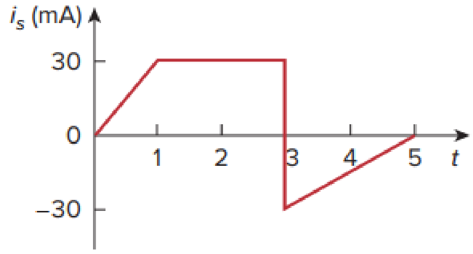
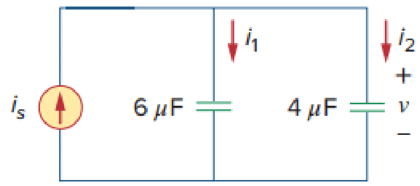
Chapter 6, Problem 31P

Problem

If $v(0) = 0$, find $v(t)$, $i1(t)$, and $i2(t)$ in the circuit of Fig. 6.63.

Fig. 6.63:





Step-by-step solution

Step 1 of 7

Draw the following circuit diagram:

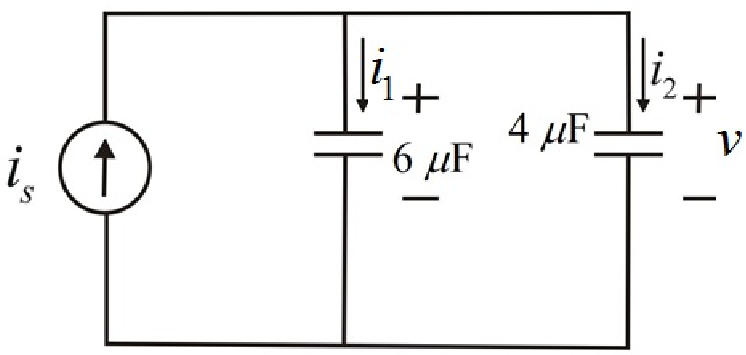


Figure 1

Step 2 of 7

Consider the current waveform is as shown in Figure 2.

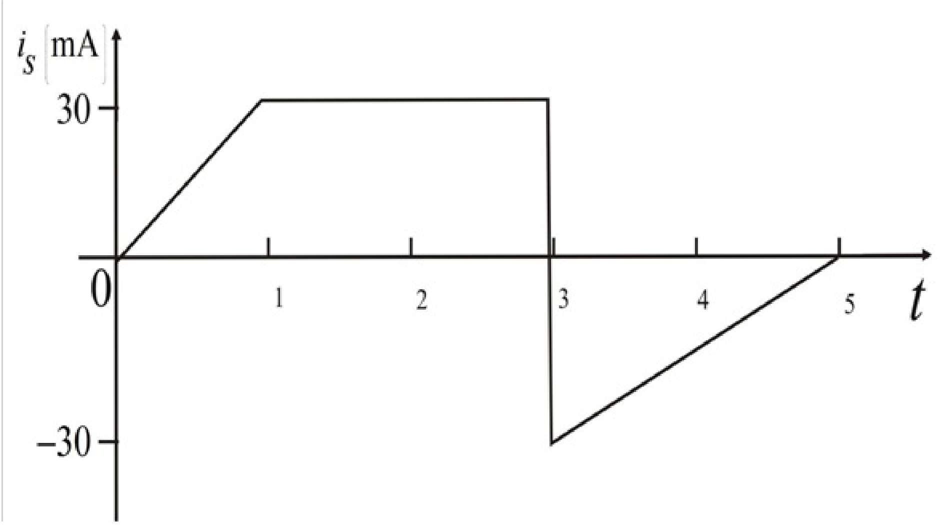


Figure 2

Step 3 of 7

From Figure 2, for $0 < t < 1$, write the line equation.

$$i_s(t) = 30t \text{ A}, 0 < t < 1$$

From Figure 2, for $1 < t < 3$, write the line equation.

$$i_s(t) = 30 \text{ A}, 1 < t < 3$$

From Figure 2, for $3 < t < 5$, write the line equation.

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y - y_1}{x - x_1}$$
$$\frac{0+30}{5-3} = \frac{y+30}{x-3}$$
$$15 = \frac{y+30}{x-3}$$
$$y = 15x - 75$$

Step 4 of 7

Write the expression for $i_s(t)$.

$$i_s(t)=\begin{cases}30t \text{ A}, & 0<t<1\\ 30 \text{ A}, & 1<t<3\\ (15t-75) \text{ A}, & 3<t<5\end{cases}$$

Calculate the value of the current $i_1(t)$.

Apply current division rule across the capacitor $6\ \mu\text{F}$ to find $i_1(t)$.

$$\begin{aligned}i_1(t)&=i_s(t)\left(\frac{6\times10^{-6}}{4\times10^{-6}+6\times10^{-6}}\right)\\&=0.6i_s(t)\end{aligned}$$

Substitute $i_s(t)$ in this expression.

$$\begin{aligned}i_1(t)&=\begin{cases}(30\times0.6)t \text{ mA}, & 0<t<1\\ (30\times0.6) \text{ mA}, & 1<t<3\\ \left[(15t-75)\times0.6\right] \text{ mA}, & 3<t<5\end{cases}\\&=\begin{cases}18t \text{ mA}, & 0<t<1\\ 18 \text{ mA}, & 1<t<3 \quad 5\\ (9t-45) \text{ mA}, & 3<t<1\end{cases}\end{aligned}$$

Therefore, the value of current $i_1(t)$ is $\boxed{\begin{cases}18t \text{ mA}, & 0<t<1\\ 18 \text{ mA}, & 1<t<3\\ (9t-45) \text{ mA}, & 3<t<5\end{cases}}$.

Step 5 of 7

Calculate the value of the current $i_2(t)$.

Apply current division rule across the capacitor $4\ \mu\text{F}$ to find $i_2(t)$.

$$i_2(t)=i_s(t)\left(\frac{4\times10^{-6}}{4\times10^{-6}+6\times10^{-6}}\right)$$
$$i_2(t)=0.4i_s(t)$$

Substitute $i_s(t)$ in this expression.

$$\begin{aligned}i_2(t)&=\begin{cases}(30\times0.4)t \text{ mA}, & 0<t<1\\ (30\times0.4) \text{ mA}, & 1<t<3\\ \left[(15t-75)\times0.4\right] \text{ mA}, & 3<t<5\end{cases}\\&=\begin{cases}12t \text{ mA}, & 0<t<1\\ 12 \text{ mA}, & 1<t<3\\ (6t-30) \text{ mA}, & 3<t<5\end{cases}\end{aligned}$$

Therefore, the value of current $i_2(t)$ is $\boxed{\begin{cases}12t \text{ mA}, & 0<t<1\\ 12 \text{ mA}, & 1<t<3\\ (6t-30) \text{ mA}, & 3<t<5\end{cases}}$.

Step 6 of 7

Calculate the value of the voltage v.

Write the expression for voltage v.

$$v = \frac{1}{C}\int i_2 dt$$

Substitute $i_2(t)$ in this expression.

$$v(t) = \begin{cases} \frac{1\times10^{-3}}{4\times10^{-6}}\int 12t \text{ V}, \ 0 < t < 1\text{s} \\ \frac{1\times10^{3}}{4}\int 12 \text{ V}, \ 1 < t < 3\text{s} \\ \frac{1}{4}\int[6t-30] \text{ kV}, \ 3 < t < 5\text{s} \end{cases}$$

$$= \begin{cases} 1.5t^2 \text{ kV}, \ 0 < t < 1\text{s} \\ [3t] \text{ kV} - v(1), \ 1 < t < 3\text{s} \\ 0.75[t^2] - 7.5[t] - v(3) \text{ kV}, \ 3 < t < 5\text{s} \end{cases}$$

$$= \begin{cases} 1.5t^2 \text{ kV}, \ 0 < t < 1\text{s} \\ [3t-15] \text{ kV}, \ 1 < t < 3\text{s} \\ 0.75t^2 - 7.5t - v(3) \text{ kV}, \ 3 < t < 5\text{s} \end{cases}$$

$$= \begin{cases} \frac{12}{4}\frac{[t^2]}{2} \text{ kV}, \ 0 < t < 1\text{s} \\ [3t-1.5] \text{ kV}, \ 1 < t < 3\text{s} \\ 0.75t^2 - 7.5t - [v(3)]\text{kV}, \ 3 < t < 5\text{s} \end{cases}$$

Step 7 of 7

Simplify voltage v further.

$$v(t) = \begin{cases} \frac{12}{4}\frac{[t^2]}{2} \text{ kV}, \ 0 < t < 1\text{s} \\ [3t-1.5] \text{ kV}, \ 1 < t < 3\text{s} \\ 0.75t^2 - 7.5t - [0.75(3^2) - 7.5(3) - [3(3)-1.5]]\text{kV}, \ 3 < t < 5\text{s} \end{cases}$$

$$= \begin{cases} \frac{12}{4}\frac{[t^2]}{2} \text{ kV}, \ 0 < t < 1\text{s} \\ [3t-1.5] \text{ kV}, \ 1 < t < 3\text{s} \\ 0.75t^2 - 7.5t[t] - [-23.25]\text{kV}, \ 3 < t < 5\text{s} \end{cases}$$

$$= \begin{cases} \frac{12}{4}\frac{[t^2]}{2} \text{ kV}, \ 0 < t < 1\text{s} \\ [3t-1.5] \text{ kV}, \ 1 < t < 3\text{s} \\ 0.75t^2 - 7.5t[t] + 23.25 \text{ kV}, \ 3 < t < 5\text{s} \end{cases}$$

Hence, the voltage across the circuit is $\boxed{\begin{cases} \frac{12}{4}\frac{[t^2]}{2} \text{ kV}, \ 0 < t < 1\text{s} \\ [3t-1.5] \text{ kV}, \ 1 < t < 3\text{s} \\ 0.75t^2 - 7.5t[t] + 23.25 \text{ kV}, \ 3 < t < 5\text{s} \end{cases}}$.